

# GigaWorld-Policy-0.5: A Faster and Stronger WAM Empowered by AutoResearch

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Project Page: <https://open-gigaai.github.io/giga-world-policy/>

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## Abstract

World Action Models (WAMs) improve robot policy learning by jointly modeling actions and future visual observations, using future scene evolution as dense supervision for physically grounded action generation. However, a common design in existing WAMs is to explicitly generate future videos at inference time, incurring substantial computational overhead and hindering real-time closed-loop deployment. GigaWorld-Policy addresses this issue with an action-centered formulation, where future visual dynamics are used during training while action-only decoding is used at inference time. Building upon this framework, we present *GigaWorld-Policy-0.5*, an enhanced action-centered WAM designed for more efficient robot control. During pretraining, *GigaWorld-Policy-0.5* adopts a mixed Action-Conditioned World Modeling (AC-WM) and WAM training strategy. This strengthens the coupling between visual dynamics and robot actions and improves the transferability of action representations for downstream policy learning. For efficient inference, *GigaWorld-Policy-0.5* introduces a Mixture-of-Transformers architecture that separates visual dynamics modeling and action generation into specialized experts, reducing active computation during action-only inference and achieving 85 ms inference latency on a local RTX 4090 setup. In addition, we employ an agent-based AutoResearch pipeline to systematically search training configurations, enabling more efficient identification of optimal experimental setups while reducing the time and manual intervention required for hyperparameter tuning. Experiments and ablations show that *GigaWorld-Policy-0.5* preserves the training benefits of future visual dynamics while improving inference efficiency for robot control.

## 1. Introduction

World models have recently made significant progress in learning predictive representations of the physical world from large-scale visual data [1, 2, 20, 24, 38, 41, 45, 46, 62, 68]. By modeling how scenes evolve over time, these models capture rich spatiotemporal priors about object motion, interaction dynamics, and long-horizon state transitions. Such predictive capability is naturally attractive for robotic control, where a policy must not only recognize the current observation, but also anticipate how its actions may change the environment.

Motivated by this progress, recent Vision-Language-Action (VLA) models have started to leverage world models to improve policy learning [8, 16, 27, 34, 36,

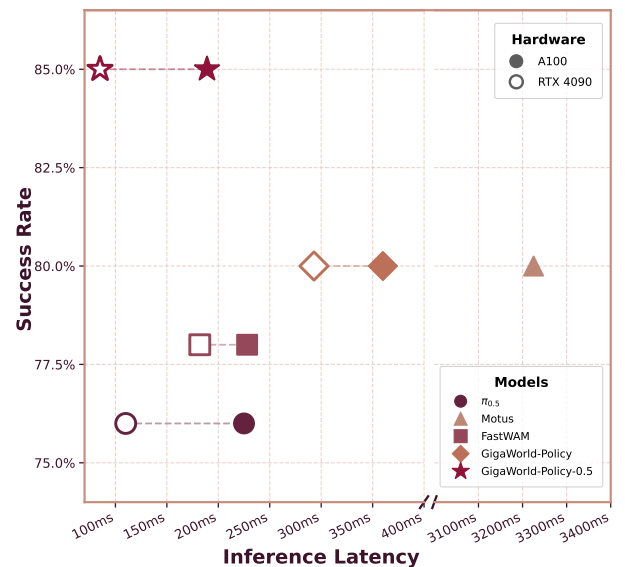


Figure 1: Comparison of *GigaWorld-Policy-0.5* with baselines on inference frequency and success rate across hardware platforms in real-world settings.

37]. Compared with prior VLA methods [5, 6, 14, 15, 31, 36, 37, 50, 53, 58] that primarily rely on sparse action supervision from robot demonstrations, world models provide complementary predictive information about how the scene may evolve. Incorporating such predictive information into VLA policies provides richer temporal context for action prediction and helps ground policy learning in future scene evolution. This line of work suggests that predictive world knowledge is an important ingredient for building more capable robot policies.

Beyond incorporating world model predictions into VLA policies, another line of work explores a more unified formulation through World Action Models (WAMs), which jointly model robot actions and future observations [4, 22, 28, 40, 54]. Instead of treating future prediction as an external input or auxiliary signal to a policy model, WAMs integrate action generation and future-dynamics modeling within the same framework. This formulation allows action representations to be learned together with their visual consequences: the model predicts not only what action should be executed, but also how the scene is expected to evolve under that action. As a result, future visual dynamics can serve as dense supervision for learning physically grounded actions.

While this unified formulation enables WAMs to acquire certain zero-shot generalization capabilities by coupling action generation with future scene evolution [56], it also introduces a practical deployment challenge. In many existing WAM approaches, the same joint modeling process used for training is also invoked during inference, requiring explicit future-video generation, iterative denoising, or predictive rollout at deployment time [33, 56]. Since video tokens are substantially more expensive than action tokens, this design introduces significant computational overhead and limits real-time closed-loop control. Moreover, errors in imagined future frames may accumulate over long horizons and weaken downstream action generation. Therefore, a key challenge is how to preserve the generalization benefit of action-dynamics coupling while avoiding costly future rollout during inference.

GigaWorld-Policy [55] addresses this challenge with an efficient action-centered World Action Model, following the broader direction of decoupling future prediction between training and inference [55, 57]. It leverages future visual dynamics to provide dense supervision during training, while allowing the policy to skip explicit future-video generation and directly decode actions at inference time. Through an action-centered causal token structure, action prediction is separated from future-video prediction, retaining the benefit of WAM training while substantially reducing inference latency for closed-loop deployment. As a result, GigaWorld-Policy demonstrates that WAMs can achieve low-latency action generation comparable to efficient VLA policies while retaining the stronger supervision and physical grounding provided by future visual dynamics.

Building upon GigaWorld-Policy, we present *GigaWorld-Policy-0.5*. *GigaWorld-Policy-0.5* inherits the core formulation of GigaWorld-Policy, where future visual dynamics are used to provide dense supervision during training, while inference can be performed through action-only decoding without explicit future-video generation. It also preserves the action-centered causal token structure: action tokens are predicted from the current observation, robot state, and language instruction, while future visual tokens are predicted conditioned on the action sequence. This masking strategy prevents future visual information from leaking into action prediction, and allows future-video prediction to remain optional during deployment, thereby retaining the low-latency control advantage of GigaWorld-Policy.

Compared with GigaWorld-Policy, *GigaWorld-Policy-0.5* introduces several updates to the model architecture and training pipeline. First, it adopts a Mixture-of-Transformers (MoT) architecture that separates visual dynamics modeling and action generation into specialized experts while preserving the action-centered WAM design. Although MoT increases the total parameter count, its expert specialization enables an action-only inference pathway that avoids unnecessary visual-dynamics computation, achieving an inference latency of approximately 85 ms on an NVIDIA RTX 4090 and thereby improving deployment efficiency. Second, during pretraining, we adopt a mixed training strategy that combines Action-Conditioned World Modeling (AC-WM) with World Action Model training. This encourages the model to better capture the relationship between visual

dynamics and robot actions, learning both general scene evolution and action-induced future changes, which in turn strengthens its action modeling capability. Finally, we employ an agent-based AutoResearch [18] pipeline to make the experimental search process more systematic and efficient. The pipeline automatically launches pilot runs, monitors validation metrics, compares candidate configurations, and promotes promising settings to longer training. Through this iterative process, AutoResearch helps derive a reliable training recipe for *GigaWorld-Policy-0.5* with reduced manual intervention.

## 2. Related Work

### 2.1. World Models as Data Engines for Robotics

Recent advances in world models [21, 30, 47, 60, 63, 65] have improved robotic video generation and prediction [11, 25, 31, 42, 64] and have increasingly served as scalable data engines or learned simulators for robot learning [3, 17, 25, 38, 44]. The central goal is to learn a generative or predictive model that captures the temporal evolution of embodied environments, enabling the synthesis or imagination of diverse visual experiences from limited real-world or simulated trajectories.

At the level of action-conditioned future prediction, Pandora [51] demonstrates that free-form text actions can be used to control video world models, enabling interactive prediction beyond passive video generation. FreeAction [19] further considers continuous robot actions and introduces action-scaled classifier-free guidance to control the intensity of generated motions. Building on such controllability, subsequent efforts have explored scalable generation and transfer of embodied visual data. RoboTransfer [25] and Cosmos-Transfer1 [3] focus on transferring visual experiences across domains, leveraging 3D or multimodal spatial conditions to improve geometric consistency and Sim2Real data generation. GigaWorld-0 [38] provides a high-fidelity video-and-3D data engine that synthesizes temporally coherent embodied trajectories with fine-grained control over appearance, scene layouts, viewpoints, and action semantics. Qwen-RobotWorld [59] further scales action-conditioned embodied video generation across multiple robotics domains, including manipulation, navigation, and driving. Going beyond video generation alone, Aether [67] unifies geometry-aware world modeling by jointly optimizing 4D dynamic reconstruction, action-conditioned video prediction, and goal-conditioned visual planning.

However, most existing efforts use world models primarily as external data engines or simulators [43, 61], while largely overlooking how their predictive priors can be embedded into deployable robot policies. This motivates recent work on world models for robot control, where future prediction and action generation are jointly considered.

### 2.2. World Models for Robot Control

World models have long been studied as predictive representations for decision making, and recent progress in large-scale video generation has renewed their relevance to robotic control. Instead of mapping observations directly to actions, world-model-based policies exploit predictions of future scene evolution to provide temporal structure, dynamics priors, or intermediate plans. Early video-based policy methods, such as UniPi [12], formulate control as future video generation followed by action recovery, while generative robot models such as GR-2 [48] further combine large-scale video pretraining with action prediction for generalizable manipulation.

World Action Models [4, 22, 33, 55], grounded in the video generation paradigm, aim to model robot actions and future visual dynamics within a unified framework. By coupling action prediction with future visual modeling, WAMs provide dense temporal supervision and learned predictive priors beyond sparse demonstration actions. VideoVLA [33] adapts pretrained video generation models into robotic manipulators, jointly predicting future visual outcomes and action sequences. Motus [4] introduces a unified latent action world model with a Mixture-of-Transformer architecture and UniDiffuser-style scheduling, enabling multiple modes including world modeling, inverse dynamics, policy learning, video generation, and joint video-action prediction. UWM [66]

similarly couples video and action diffusion within a unified framework, showing that heterogeneous video and robot data can support scalable policy learning. Building on this direction, MotuBrain [40] extends Motus with multiview modeling, cross-embodiment action representations, and deployment-oriented optimization for real-time control.

Another branch investigates how much future prediction must be explicitly realized during policy inference. Mimic-video [32] uses an Internet-scale video backbone to produce latent video plans, which are decoded into actions through a flow-matching inverse-dynamics model. LingBot-VA [22] adopts autoregressive video-action world modeling with closed-loop rolling prediction to mitigate error accumulation during long-horizon execution. S-VAM [52] distills multi-step video foresight into efficient geometric and semantic representations, DiT4DiT [29] uses intermediate denoising features rather than decoded frames for action prediction, and LaWAM [9] exposes future dynamics through compact latent visual subgoals instead of pixel-level video.

Complementary to these approaches, GigaWorld-Policy [55] and Fast-WAM [57] explore more efficient ways to incorporate predictive world knowledge into action policies. Both methods use video-based predictive supervision during training while avoiding the need to explicitly generate future videos at test time. Together, these works indicate that the benefits of world modeling need not rely on computationally expensive pixel-level future prediction during policy execution.

Overall, prior work has explored a spectrum of designs that reduce the reliance on explicit pixel-level future rollout during policy inference. However, achieving the low latency required for closed-loop control on edge devices remains challenging. This motivates *GigaWorld-Policy-0.5*, which retains an action-centered world action modeling formulation while improving inference efficiency through a lightweight action-only execution pathway.

### 3. Method

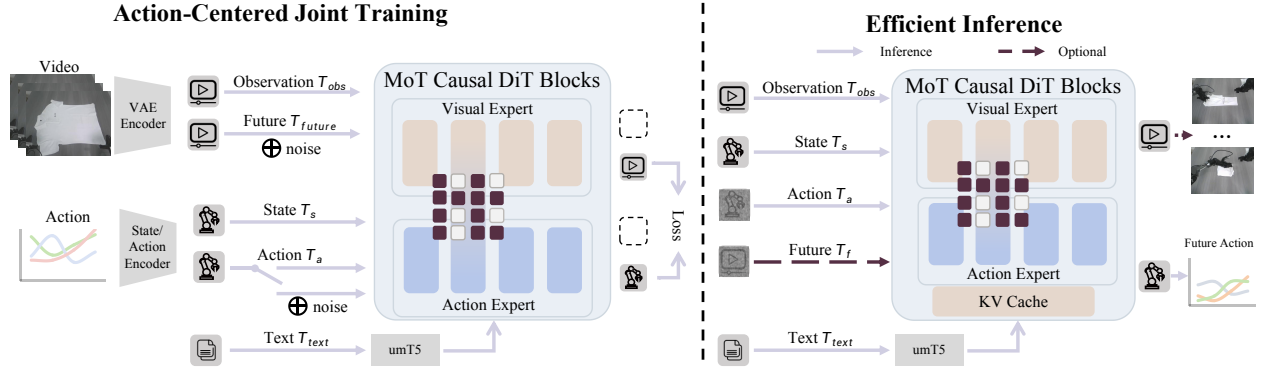


Figure 2: Overview of *GigaWorld-Policy-0.5*, an MoT-based action-centered World Action Model. The model consists of a visual expert and an action expert: the visual expert specializes in processing video tokens, while the action expert focuses on action-token modeling. The two experts are connected through multi-modal self-attention, which follows the same causal masking strategy as *GigaWorld-Policy* [55] to preserve action-centered dependency modeling.

#### 3.1. Overview of *GigaWorld-Policy-0.5*

The overall framework of *GigaWorld-Policy-0.5* is illustrated in Fig. 2. It follows the formulation of the action-centered world action model introduced in *GigaWorld-Policy* [55]. The policy prediction problem under this action-centered setting can be formulated as follows. Given the current multi-view observation  $\{o_t^{\text{left}}, o_t^{\text{front}}, o_t^{\text{right}}\}$ , proprioceptive state  $s_t$ , and language instruction  $l$ , the model jointly predicts an action

chunk  $\hat{\mathbf{a}}_{t:t+p-1}$  of length  $p$  and future visual observations  $\hat{\mathbf{o}}_{t+\Delta:t+K\Delta}^{\text{comp}}$ :

$$(\hat{\mathbf{a}}_{t:t+p-1}, \hat{\mathbf{o}}_{t+\Delta:t+K\Delta}^{\text{comp}}) \sim g_{\theta}(\cdot | o_t^{\text{comp}}, s_t, l), \quad (1)$$

where

$$\hat{\mathbf{a}}_{t:t+p-1} = (\hat{a}_t, \hat{a}_{t+1}, \dots, \hat{a}_{t+p-1}), \quad \hat{\mathbf{o}}_{t+\Delta:t+K\Delta}^{\text{comp}} = (\hat{o}_{t+\Delta}^{\text{comp}}, \hat{o}_{t+2\Delta}^{\text{comp}}, \dots, \hat{o}_{t+K\Delta}^{\text{comp}}). \quad (2)$$

Here,  $\Delta$  denotes the temporal stride for future visual prediction, and  $K = \lfloor p/\Delta \rfloor$  denotes the number of predicted future observations within the action horizon.

The composite observation  $o^{\text{comp}}$  is constructed by concatenating the left, front, and right camera views:

$$o^{\text{comp}} = \text{Compose}(o^{\text{left}}, o^{\text{front}}, o^{\text{right}}). \quad (3)$$

Notably, *GigaWorld-Policy-0.5* adopts a similar multi-view composition strategy as Motus [4], where the front view is placed on the top and the left and right views are concatenated in the bottom row to form a unified image input for visual encoding.

To better transfer world-model priors into policy learning, *GigaWorld-Policy-0.5* adopts the same causal masking strategy as GigaWorld-Policy [55]. Specifically, future visual tokens are allowed to attend to action tokens, while action tokens are prevented from attending to future visual tokens. This design has two benefits. First, during training, the causal attention pattern implicitly conditions future visual prediction on the action tokens, encouraging the model to learn action representations that are consistent with the world-model prior and the expected scene evolution. Second, during inference, future visual prediction is optional; when low-latency control is desired, the model can skip the large number of future visual tokens and directly decode action tokens.

*GigaWorld-Policy-0.5* is optimized with the flow matching framework [13, 23]. For action tokens and future visual tokens, we sample modality-specific flow timesteps with different flow-shift factors. Given  $r^a, r^v \sim \mathcal{U}(0, 1)$  and flow-shift factors  $\gamma_a$  and  $\gamma_v$ , the shifted timesteps are computed as

$$\tau^a = \frac{\gamma_a r^a}{1 + (\gamma_a - 1)r^a}, \quad \tau^v = \frac{\gamma_v r^v}{1 + (\gamma_v - 1)r^v}. \quad (4)$$

We then form a joint timestep vector for the action and future visual modalities:

$$\tau = \begin{bmatrix} \tau^a \\ \tau^v \end{bmatrix}. \quad (5)$$

Let  $x_1^a$  and  $x_1^v$  denote the clean action tokens and future visual tokens, respectively, and let  $x_0^a$  and  $x_0^v$  denote their corresponding Gaussian noise. The noisy joint token is constructed by linearly interpolating between noise and data for each modality:

$$x_{\tau} = \tau \begin{bmatrix} x_1^a \\ x_1^v \end{bmatrix} + (1 - \tau) \begin{bmatrix} x_0^a \\ x_0^v \end{bmatrix}. \quad (6)$$

The corresponding ground-truth velocity is given by

$$\nu_{\tau} = \frac{dx_{\tau}}{d\tau} = \begin{bmatrix} x_1^a - x_0^a \\ x_1^v - x_0^v \end{bmatrix}. \quad (7)$$

Finally, *GigaWorld-Policy-0.5* is optimized by regressing the model-predicted velocity field to the ground-truth

flow velocity:

$$\mathcal{L} = \mathbb{E}_{x_1, x_0, \tau} \left[ \|g_\theta(x_\tau, \tau \mid o_t^{\text{comp}}, s_t, l) - \nu_\tau\|_2^2 \right]. \quad (8)$$

### 3.2. Model Architecture

Fig. 2 illustrates the MoT-based architecture of *GigaWorld-Policy-0.5*. The model takes visual observations, proprioceptive states, action chunks, and language instructions as inputs. For visual inputs, we use the visual VAE from Wan [41] to encode the composite observations  $o^{\text{comp}}$  into visual latent tokens. For non-visual inputs, robot states and action chunks are mapped to the visual hidden dimension via multi-layer perceptrons (MLPs), producing state tokens and action tokens. The language instruction is encoded by umT5 [10], whose text embeddings are used as conditioning signals for instruction-following control.

Given these tokens, *GigaWorld-Policy-0.5* follows the action-centered world action framework but replaces the fully shared Transformer backbone with an MoT structure. The MoT architecture separates the two central modeling components of action-centered WAMs, namely visual dynamics modeling and action generation, into a visual expert and an action expert. The visual expert processes current and future visual tokens and models scene evolution in the latent visual space, while the action expert focuses on action-token denoising and action-chunk prediction. Each expert is equipped with its own cross-attention and feed-forward network (FFN) modules, enabling modality-specific language conditioning and nonlinear transformation. These two experts are connected through multi-modal self-attention, enabling information exchange across visual, state, and action tokens while preserving a unified token sequence for joint world action modeling.

In particular, the multi-modal self-attention module follows *GigaWorld-Policy* [55] by adopting an action-centered causal mask to regulate cross-modal information flow. Specifically, action tokens are allowed to attend to the current visual tokens, state tokens, and language conditioning, but are prevented from attending to future visual tokens. Future visual tokens, on the other hand, can attend to the current context and action tokens, enabling action-conditioned future visual prediction. This causal attention mask prevents information leakage from future observations into action prediction, while allowing future visual dynamics to provide dense training-time supervision. At inference time, future visual tokens can be omitted, and the model directly decodes action tokens for low-latency control.

For parameter initialization, *GigaWorld-Policy-0.5* initializes the visual expert with pretrained visual weights from *GigaWorld-1* [39], a large-scale pretrained world model. This provides the visual expert with a world-model prior for visual dynamics modeling, allowing the model to inherit general knowledge about scene evolution before being adapted to robot-specific observations and interactions. For the newly introduced action expert, we initialize its parameters from the corresponding visual-expert weights. For parameters whose dimensions are not aligned, we initialize the action-expert parameters by taking the leading  $n$  dimensions of the corresponding visual-expert weights, where  $n$  denotes the target dimensionality of the action-expert parameter.

### 3.3. Training Pipeline

The training pipeline of *GigaWorld-Policy-0.5* consists of two stages: robot-data pretraining and target-robot post-training. Before these two stages, the visual expert is initialized from *GigaWorld-1* [39], which is pretrained on over ten thousand hours of video data. This initialization provides *GigaWorld-Policy-0.5* with a general world-model prior for visual dynamics modeling.

In the pretraining stage, we train *GigaWorld-Policy-0.5* on 2K hours of filtered open-source robot data [7, 26, 35, 49] and internally collected real-robot data. This stage adapts the pretrained world-model prior to robot-centric scenarios, including robot embodiments, manipulation workspaces, camera viewpoints, and interaction patterns. The model learns to predict future visual evolution under robot-relevant observations and actions, improving its alignment with robotic control settings before target-domain specialization. Notably, to better model the



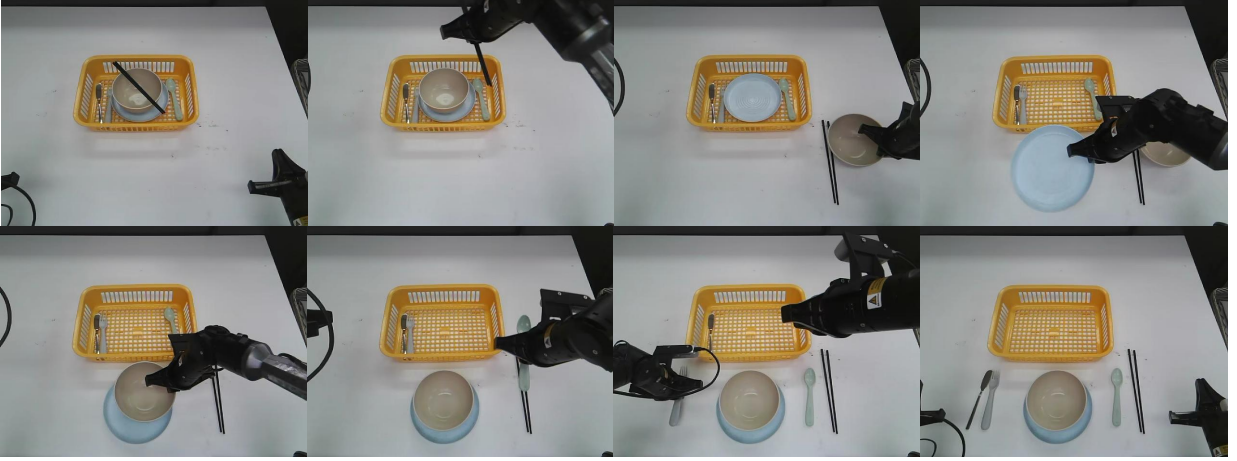


Figure 3: Real-world demonstration of *GigaWorld-Policy-0.5* on the *Tableware Arrangement* task.

relationship between actions and visual dynamics, we also incorporate action-conditioned world-model training during this stage, where future visual evolution is predicted under robot-relevant observations and actions. This encourages the model to learn how robot actions affect scene changes before target-domain specialization.

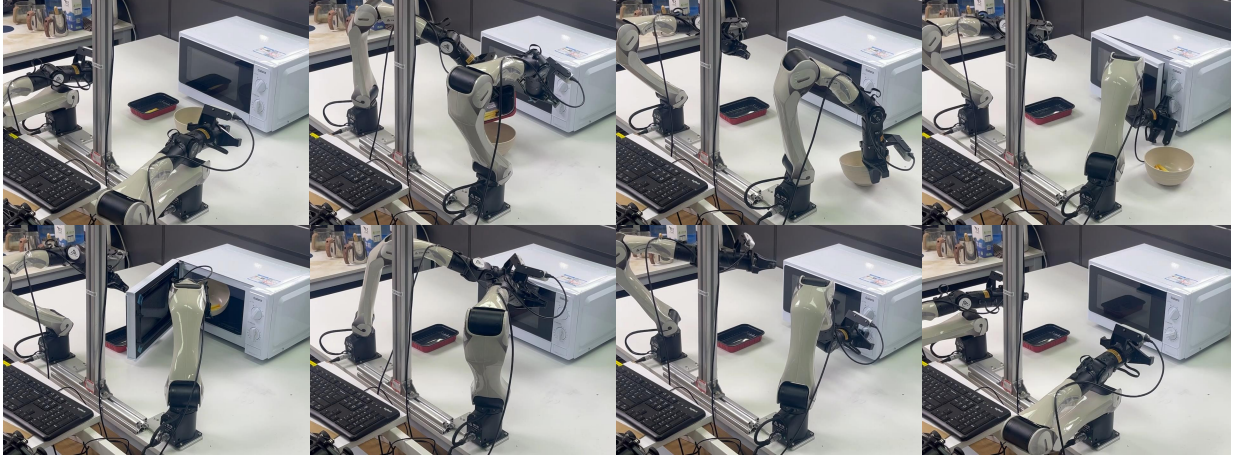
In the post-training stage, *GigaWorld-Policy-0.5* is trained on target real-robot trajectories that contain aligned observations, language instructions, robot states, and action sequences. The model jointly optimizes action prediction and future visual dynamics modeling under the action-centered causal structure. The action objective teaches the model to generate instruction-conditioned action chunks for closed-loop control, while the future-visual objective provides auxiliary supervision that encourages action representations to remain consistent with plausible scene evolution. During deployment, we use the action-only inference path: future visual tokens are not generated, and the model directly decodes actions conditioned on the current observation, state, and instruction.

### 3.4. Inference Acceleration

To support real-world deployment on edge computing platforms, we optimize the inference stack through KV caching, graph compilation, and a lightweight C++ runtime. During autoregressive action generation, the visual and language context remains unchanged across decoding steps; we therefore cache the key-value states of all attention layers after encoding the input context and reuse them for subsequent action-token predictions, avoiding redundant attention computation. We further apply `torch.compile` to optimize the inference graph, which reduces Python-level dispatch overhead and enables backend-level optimizations such as operator fusion and more efficient memory scheduling. Finally, we deploy the optimized policy in a C++ runtime that integrates image preprocessing, tensor construction, model execution, KV-cache management, and action post-processing into a unified native pipeline. This design removes the Python dependency, reduces runtime overhead, and simplifies integration with robot-control middleware, enabling low-latency closed-loop policy execution on edge devices.

## 4. Experiments

**Evaluation Metrics.** We primarily report the Success Rate (SR) to evaluate task completion performance. For real-world experiments, we evaluate gripper-based manipulation under two settings: **text following** and **long-horizon task** execution. For **text following**, we use a graded SR to better capture partial task completion: each trial is scored by four equally weighted stages, with 0.25 assigned to each stage, including reaching the target object, grasping it, moving to the target location, and successfully placing it. For **long-horizon tasks**, SR

Figure 4: Real-world demonstration of *GigaWorld-Policy-0.5* on the *Food Heating* task.

follows the same binary definition as in simulation, where  $SR = 1$  only if the full task sequence is completed and  $SR = 0$  otherwise.

**Baselines.** We compare *GigaWorld-Policy-0.5* with several representative baselines spanning two major paradigms. For VLM-based VLA methods, we include  $\pi_{0.5}$  [15], which uses a large vision-language backbone to align visual observations with language instructions and decode low-level control actions through an action head. For WAM-based methods, we include Motus [4], FastWAM [57], and *GigaWorld-Policy* [55], which incorporate visual dynamics or world-modeling objectives to support policy learning.

**Implementation Details.** We use the pretrained *GigaWorld-1* [39] to initialize the visual expert, enabling the model to inherit strong visual generative representations. The action expert is then initialized from the visual expert by copying the corresponding weights; for layers with mismatched dimensions, we retain the compatible submatrices and truncate the remaining entries to match the action-expert configuration. In our implementation, the visual expert uses a hidden dimension of 3072 and an FFN dimension of 14336, while the action expert is made more lightweight with a hidden dimension of 1024 and an FFN dimension of 4096. For the action chunk length and future-observation stride, we follow the settings adopted in *GigaWorld-Policy*.

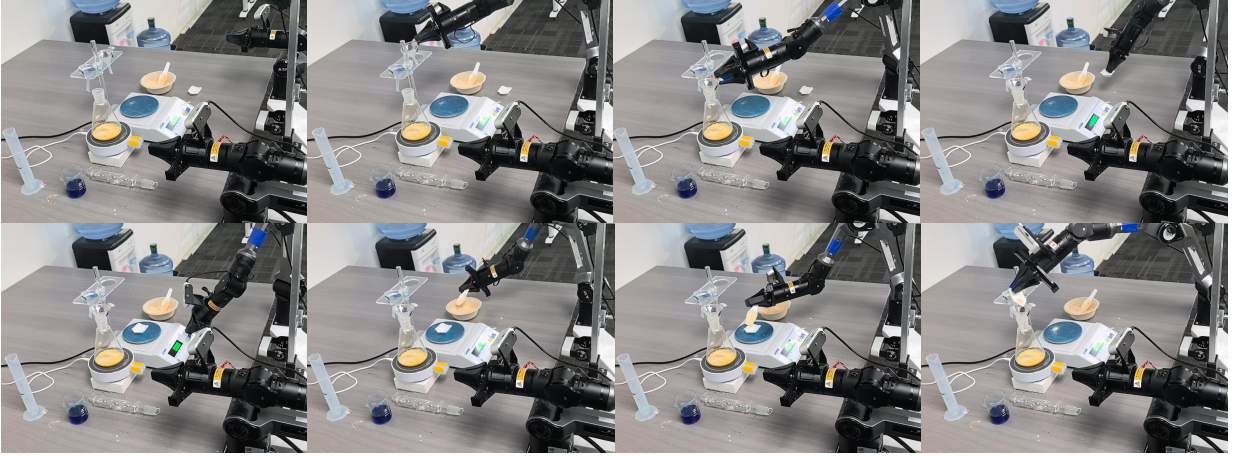
Table 1: Evaluation of text-following ability on the fruit-picking task.

| Task Name      | Text Instruction                                         | $\pi_{0.5}$ [15] | Motus [4] | FastWAM [57] | <i>GigaWorld-Policy</i> [55] | <i>GigaWorld-Policy-0.5</i> |
|----------------|----------------------------------------------------------|------------------|-----------|--------------|------------------------------|-----------------------------|
| Pick the Fruit | Pick the <i>banana</i> and place it into the basket.     | 0.88             | 0.93      | 0.83         | 0.90                         | <b>0.95</b>                 |
|                | Pick the <i>apple</i> and place it into the basket.      | 0.73             | 0.78      | 0.75         | 0.73                         | <b>0.80</b>                 |
|                | Pick the <i>lemon</i> and place it into the basket.      | 0.68             | 0.75      | 0.73         | 0.78                         | <b>0.83</b>                 |
|                | Pick the <i>grape</i> and place it into the basket.      | 0.85             | 0.83      | 0.88         | 0.88                         | <b>0.93</b>                 |
|                | Pick the <i>avocado</i> and place it into the basket.    | 0.68             | 0.70      | 0.75         | 0.73                         | <b>0.78</b>                 |
|                | Pick the <i>strawberry</i> and place it into the basket. | 0.78             | 0.80      | 0.73         | 0.78                         | <b>0.85</b>                 |
| Average        | –                                                        | 0.76             | 0.80      | 0.78         | 0.80                         | <b>0.85</b>                 |

#### 4.1. Real-World Results

We conduct gripper-based manipulation experiments on an AgileX PiPER 6-DoF robotic arm from two complementary perspectives. **Text following** evaluates language grounding by executing multiple instructions under the same scene configuration. Each trial is scored using four equally weighted stages (0.25 point each): reaching the target object, grasping the target object, moving to the target placement location, and successfully placing the object. The final score is averaged over 10 real-world trials. **Long-horizon tasks** evaluate end-to-end sequential manipulation, where each task is executed 10 times on the real robot and only the overall task success rate is reported.



Figure 5: Real-world demonstration of *GigaWorld-Policy-0.5* on the *Solid Weighing* task.

Tab. 1 and 2 evaluate text-following ability under variations in object semantics and target locations. On the fruit-picking task, *GigaWorld-Policy-0.5* achieves an average success rate of 0.85, outperforming  $\pi_{0.5}$  [15], Motus [4], FastWAM [57], and GigaWorld-Policy [55] by 0.09, 0.05, 0.07, and 0.05, respectively. It consistently obtains the highest success rate across all six fruit categories, with particularly clear gains on lemon and avocado, suggesting that the model can reliably ground fine-grained language descriptions to visually distinct objects. On the more compositional object-placement task, *GigaWorld-Policy-0.5* achieves an average success rate of 0.89, outperforming the strongest baseline, Motus, by 0.06. This task requires jointly identifying the referenced object and following the specified destination relation (e.g., placing an object *on* a plate or *into* a basket). Our method attains the best result on every instruction and shows notable improvements for bowl-to-basket and fork-to-basket instructions, indicating stronger grounding of both object identities and spatial goal specifications. Overall, the results demonstrate that *GigaWorld-Policy-0.5* follows diverse natural-language instructions more reliably than prior methods, particularly when task execution requires precise object selection and compositional object-destination reasoning.

Table 2: Evaluation of text-following ability on the object-placement task.

| Task Name     | Text Instruction                                               | $\pi_{0.5}$ [15] | Motus [4] | FastWAM [57] | GigaWorld-Policy [55] | <i>GigaWorld-Policy-0.5</i> |
|---------------|----------------------------------------------------------------|------------------|-----------|--------------|-----------------------|-----------------------------|
| Place Objects | Pick up the <i>bowl</i> and place it on the <i>plate</i> .     | 0.87             | 0.88      | 0.73         | 0.80                  | <b>0.93</b>                 |
|               | Pick up the <i>fork</i> and place it on the <i>plate</i> .     | 0.78             | 0.83      | 0.85         | 0.80                  | <b>0.88</b>                 |
|               | Pick up the <i>spoon</i> and place it on the <i>plate</i> .    | 0.73             | 0.83      | 0.80         | 0.75                  | <b>0.85</b>                 |
|               | Pick up the <i>bowl</i> and place it into the <i>basket</i> .  | 0.75             | 0.83      | 0.80         | 0.88                  | <b>0.95</b>                 |
|               | Pick up the <i>fork</i> and place it into the <i>basket</i> .  | 0.65             | 0.75      | 0.68         | 0.78                  | <b>0.85</b>                 |
|               | Pick up the <i>spoon</i> and place it into the <i>basket</i> . | 0.80             | 0.85      | 0.75         | 0.83                  | <b>0.88</b>                 |
| Average       | –                                                              | 0.76             | 0.83      | 0.77         | 0.81                  | <b>0.89</b>                 |

Table 3 reports end-to-end success rates on three long-horizon manipulation tasks, each of which requires the policy to execute multiple temporally dependent steps. *GigaWorld-Policy-0.5* achieves the best performance on every task, attaining an average success rate of 0.77. Compared with the strongest baseline, Motus [4], it yields an absolute improvement of 0.20 and a relative improvement of 35%. This substantial gain highlights the effectiveness of our method in long-horizon task execution, where successful completion depends on maintaining coherent action sequences and accurately handling intermediate state transitions. The results further suggest that the predictive representations acquired through world modeling provide useful temporal and interaction-aware information for long-horizon robotic manipulation.

Table 3: Evaluation of end-to-end success rates on long-horizon tasks.

| Task Name             | $\pi_{0.5}$ [15] | Motus [4] | FastWAM [57] | GigaWorld-Policy [55] | <i>GigaWorld-Policy-0.5</i> |
|-----------------------|------------------|-----------|--------------|-----------------------|-----------------------------|
| Food Heating          | 0.50             | 0.60      | 0.50         | 0.60                  | <b>0.80</b>                 |
| Solid Weighing        | 0.30             | 0.50      | 0.40         | 0.40                  | <b>0.70</b>                 |
| Tableware Arrangement | 0.70             | 0.60      | 0.50         | 0.60                  | <b>0.80</b>                 |
| Average               | 0.50             | 0.57      | 0.47         | 0.53                  | <b>0.77</b>                 |

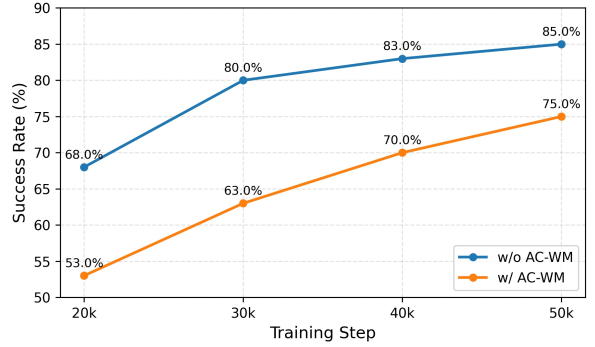
## 4.2. Ablation Studies

We conduct two groups of ablation studies to analyze *GigaWorld-Policy-0.5*. The first group focuses on standard component ablations, where we isolate the contribution of key design choices in the model and training pipeline. The second group uses the AutoResearch [18] pipeline to study the effect of hyperparameter choices, such as learning rate and warmup strategy, under a unified evaluation protocol.

### Effect of the mixed AC-WM and WAM pretraining.

We evaluate whether incorporating Action-Conditioned World-Model (AC-WM) training during pretraining improves downstream policy learning. This ablation compares two pretraining settings: one using standard WAM pretraining only, and the other mixing AC-WM and WAM pretraining, where future visual observations are predicted conditioned on robot actions. After pretraining, both models are post-trained using the same policy training recipe and evaluated on the *pick the fruit* task.

Fig. 6 shows that the model pretrained with mixed AC-WM and WAM outperforms the baseline throughout post-training: it converges faster, attains higher success



rates, and ultimately reaches a success rate of 0.85. Notably, strong performance emerges at substantially earlier training steps, indicating that explicitly modeling how robot actions drive visual state transitions yields more transferable action representations. As a result, downstream policy learning becomes more sample-efficient, requiring fewer post-training updates to achieve competitive closed-loop control performance.

### Effect of MoT architecture.

We evaluate the effect of the MoT architecture on inference efficiency. The MoT architecture processes action generation and video dynamics modeling with separate experts. In our design, we reduce the dimensional configuration of the action expert relative to the video expert, making the action pathway more lightweight than the video modeling pathway. As a result, although *GigaWorld-Policy-0.5* introduces additional parameters due to the expert-based architecture, the computation required for action-only inference is reduced. In addition, this expert-separated design makes it easier to initialize the video expert from pretrained video generation models, which provides stronger visual dynamics representations and further improves downstream policy performance.

As shown in Tab. 4, this architectural design leads to clear inference efficiency gains. With KV cache and torch compilation enabled, *GigaWorld-Policy-0.5* achieves an inference latency of 189 ms on an A100 GPU. Compared with FastWAM under the same torch-compiled setting, *GigaWorld-Policy-0.5* reduces latency from 229 ms to 189 ms, corresponding to a 17.5% speedup. It also outperforms the VLA-based  $\pi_{0.5}$  [15] in inference efficiency, reducing latency from 225 ms to 189 ms. Beyond the A100 setting, *GigaWorld-Policy-0.5* further achieves 110 ms latency on a local RTX 4090 edge-side setup, matching the inference speed of  $\pi_{0.5}$ . With a C++ deployment, latency can be further reduced to 85 ms, 23% faster than  $\pi_{0.5}$  and 53% faster than FastWAM [57],

Table 4: Comparison of inference efficiency and real-robot performance.

| Method                      | Latency on A100 (ms) ↓ | Latency on RTX 4090 (ms) ↓ | SR on Real-Robot ↑ |
|-----------------------------|------------------------|----------------------------|--------------------|
| $\pi_{0.5}$ [15]            | 225                    | 110                        | 0.76               |
| Motus [4]                   | 3231                   | -                          | 0.80               |
| FastWAM [57]                | 229                    | 182                        | 0.78               |
| GigaWorld-Policy [55]       | 360                    | 293                        | 0.80               |
| <i>GigaWorld-Policy-0.5</i> | 189                    | 110                        | <b>0.85</b>        |
| w/ C++ deployment           | <b>140</b>             | <b>85</b>                  | <b>0.85</b>        |

demonstrating its potential for efficient real-world deployment.

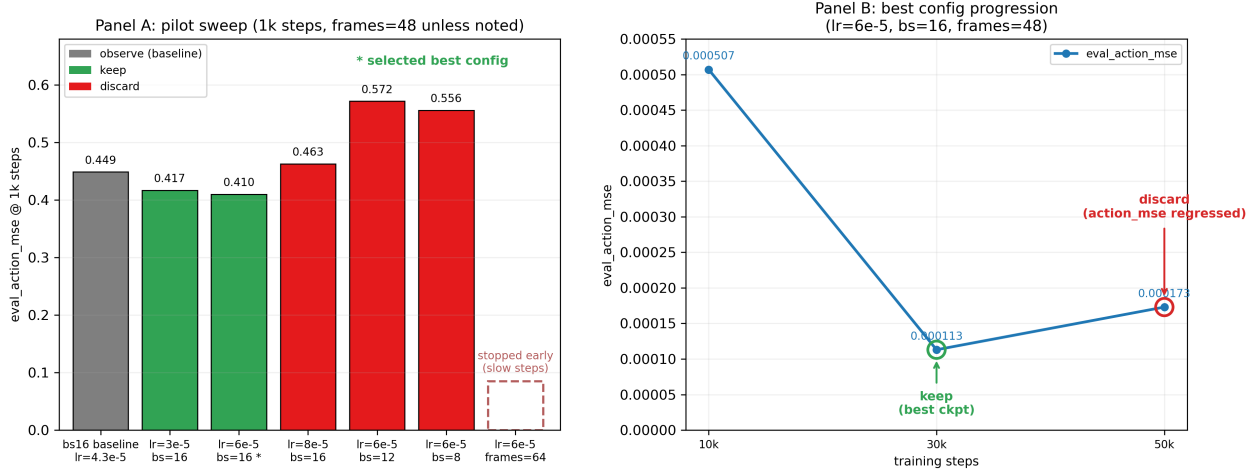


Figure 7: AutoResearch hyperparameter search and training progression on the *pick the fruit* task. Left: 1K-step pilot sweep over learning rate and batch-size configurations, where green bars indicate retained candidates and red bars indicate discarded configurations. Right: extended training progression under the selected configuration, showing that the model achieves the best validation action MSE at 30K steps.

#### AutoResearch-driven hyperparameter study.

We use AutoResearch [18] to systematically explore the training hyperparameters of *GigaWorld-Policy-0.5*, including the learning rate and warmup schedule. AutoResearch automates the full optimization loop, including candidate configuration generation, pilot training, validation metric collection, candidate selection, and extended training. This procedure is particularly useful for *GigaWorld-Policy-0.5*, as the model jointly optimizes action prediction and future visual dynamics modeling, and different hyperparameter choices may affect these two objectives in different ways.

We conduct the study on the *pick the fruit* task using approximately 3.9 hours of robot demonstration data. The dataset contains 930 episodes, of which 300 episodes are used for training and 30 episodes are held out for validation. All candidate runs use the same model initialization, data split, optimizer, and evaluation protocol. To improve the efficiency of hyperparameter exploration, AutoResearch first uses short 1K-step pilot runs to rapidly iterate over different candidate configurations. After identifying promising hyperparameters, AutoResearch performs longer training runs under the selected configuration for final model selection.

As shown on the left side of Fig. 7, AutoResearch first sweeps over candidate learning rates using these 1K-step pilot runs. The detailed learning-rate ablation results are reported in Tab. 5: among the tested candidates,  $6 \times 10^{-5}$  achieves the lowest training action loss of 0.252476 and the best evaluation action MSE of 0.409764, while  $3 \times 10^{-5}$  obtains the lowest training visual loss of 0.172330. Since action prediction quality is more directly related to downstream policy execution, AutoResearch selects  $6 \times 10^{-5}$  as the learning rate for the

Table 5: AutoResearch learning-rate sweep. Each candidate is trained for 1K steps under the same setting.

| Learning Rate          | Train Action Loss ↓ | Train Visual Loss ↓ | Eval Action MSE ↓ |
|------------------------|---------------------|---------------------|-------------------|
| $3 \times 10^{-5}$     | 0.257300            | <b>0.172330</b>     | 0.416832          |
| $4.316 \times 10^{-5}$ | 0.256593            | 0.172893            | 0.449387          |
| $6 \times 10^{-5}$     | <b>0.252476</b>     | 0.173318            | <b>0.409764</b>   |
| $8 \times 10^{-5}$     | 0.261832            | 0.175986            | 0.461381          |

next-stage search.

AutoResearch then fixes this learning rate and conducts an additional batch-size sweep. The batch-size search shows that alternative batch sizes do not outperform the original setting, so AutoResearch retains the original batch size of 16. Based on these results, AutoResearch selects  $6 \times 10^{-5}$  and a batch size of 16 as the final hyperparameter configuration. After fixing these hyperparameters, AutoResearch further scales up the training by extending the number of training steps. As shown on the right side of Fig. 7, the model reaches its best validation performance at 30K training steps. We therefore use the 30K-step checkpoint for subsequent real-robot evaluation.

## 5. Conclusion

In this technical report, we introduced GigaWorld-Policy-0.5, an enhanced action-centered World Action Model built upon GigaWorld-Policy. GigaWorld-Policy-0.5 preserves the key principle of using future visual dynamics as training-time supervision while enabling action-only decoding during inference. On top of this formulation, the model adopts a Mixture-of-Transformers architecture to separate visual dynamics modeling from action generation, reducing active computation for low-latency deployment. We also introduce a mixed pretraining strategy that combines action-conditioned world modeling with WAM training, encouraging the model to better capture action-induced visual changes and improving action modeling. Finally, an agent-based AutoResearch pipeline is used to make hyperparameter search more systematic and efficient. Overall, GigaWorld-Policy-0.5 demonstrates a practical path toward faster and more deployable World Action Models for robot control. With C++ deployment on a local RTX 4090, it further achieves 85 ms inference latency, improving the efficiency-performance trade-off of action-centered WAMs by retaining dense supervision from future visual dynamics while reducing the cost of action generation at inference time.

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